Complex Numbers ...

General n-th root of a complex number

We want to solve $\boxed{z^n = w}$.

Find all complex numbers z whose n-th power is w.

Let $w = \rho e^{j\phi}$ and look for solutions of the form $z = r e^{j\theta}$.

Then $z^n = r^n e^{jn\theta}$, so we want $\underbrace{r^n e^{jn\theta}}_{z^n} = \underbrace{\rho e^{j\phi}}_{w}$.

But the angles are not unique, so more precisely

$$r^n e^{jn\theta} = \rho e^{j(\phi + 2k\pi)} \quad k \in \mathbb{Z}$$

Therefore we must have

$$r^n = \rho \quad \text{and} \quad n\theta = \phi + 2k\pi$$

So:

$$r = \rho^{1/n} \quad \text{and} \quad \theta = \frac{\phi + 2k\pi}{n}$$

Hence the solutions are:

$$z_k = \rho^{1/n} e^{j\left(\frac{\phi + 2k\pi}{n}\right)} \quad k = 0, 1, 2, 3, \dots, n-1$$

(*) Special case if $w = 0 \longrightarrow z = 0$

example:

Solve $z^3 = 8$.

We write 8 in exponential form: $8 = 8\, e^{j \cdot 0}$

But because angles repeat $8 = 8\, e^{j(0+2k\pi)}$ $k = 0, 1, 2$

Now we apply the formula above:

$$r = 8^{1/3} = 2$$

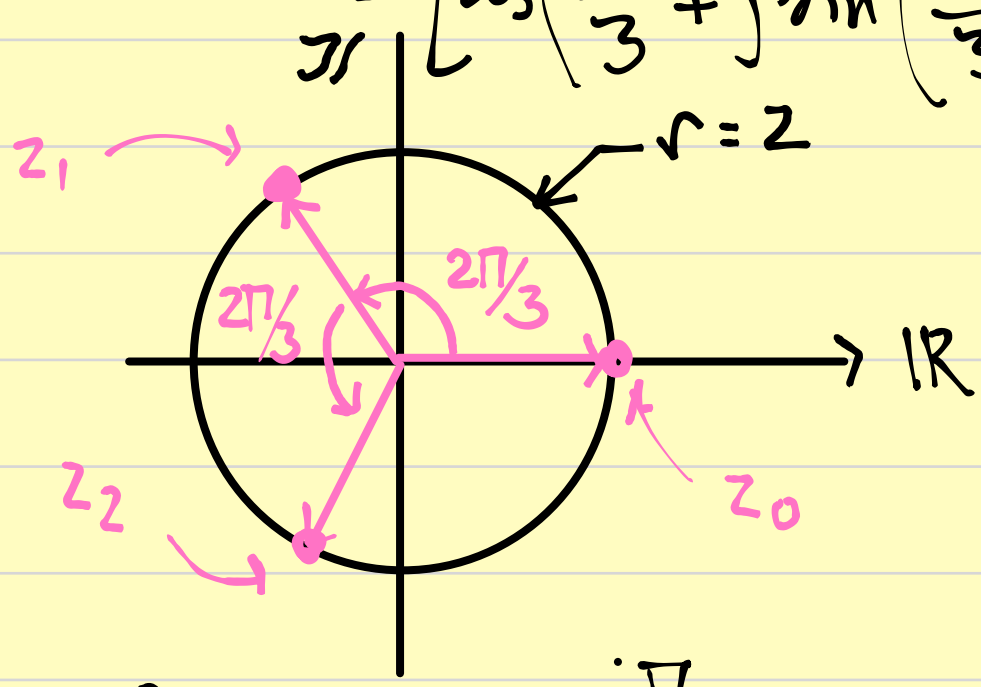
Angles: $\theta_k = \dfrac{0 + 2k\pi}{3}$

$$z_0 = 2\, e^{j0} = 2$$

$$z_1 = 2\, e^{j\frac{2\pi}{3}} = 2 \cdot \left[\cos\left(\frac{2\pi}{3}\right) + j \sin\left(\frac{2\pi}{3}\right)\right] = -1 + \sqrt{3}\, j$$

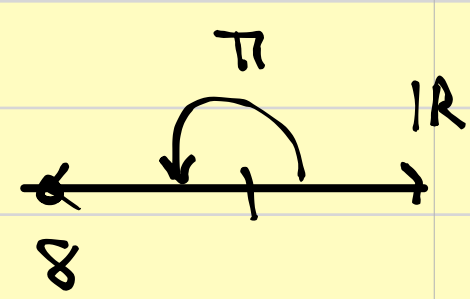
$$z_2 = 2\, e^{j\frac{4\pi}{3}} = 2 \cdot \left[\cos\left(\frac{2\pi}{3}\right) + j \sin\left(\frac{2\pi}{3}\right)\right] = -1 - \sqrt{3}\, j$$

$$\boxed{z^3 = 8}$$



exercise $z^3 = -8 = 8\, e^{j\pi}$

$$-8 = 8\, e^{j(\pi + 2k\pi)}$$



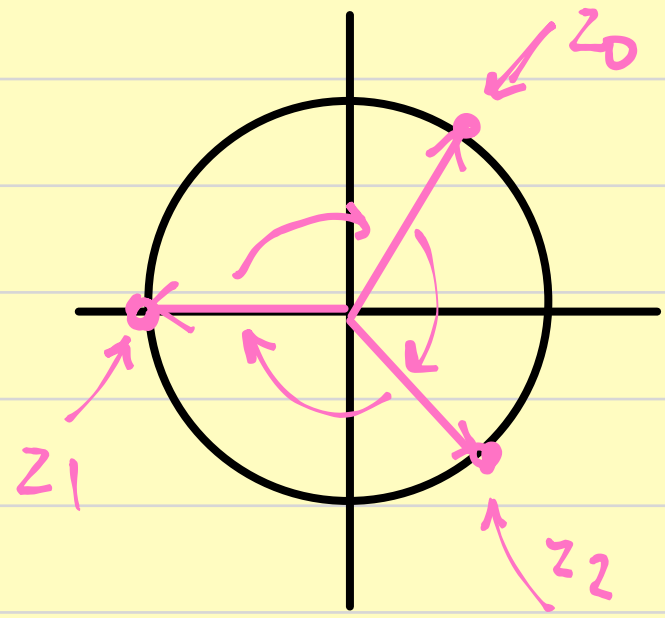
$r = \sqrt[3]{8} = 2$

Angles: $\theta_k = \dfrac{\pi + 2k\pi}{3}$

$z_0 = 2e^{j\frac{\pi}{3}} = 1 + \sqrt{3}\,j$

$z_1 = 2e^{j\pi} = -2$

$z_2 = 2e^{j\frac{5\pi}{3}} = 1 - \sqrt{3}\,j$



The most important special case $z^n = 1$

Since $1 = e^{j \cdot 2k\pi}$ the roots are $z_k = e^{j\frac{2k\pi}{n}}$ $k = 0, 1, \dots, n-1$

Geometric meaning: All roots lie on the unit circle and are equally spaced. The angle between roots is $\dfrac{2\pi}{n}$.

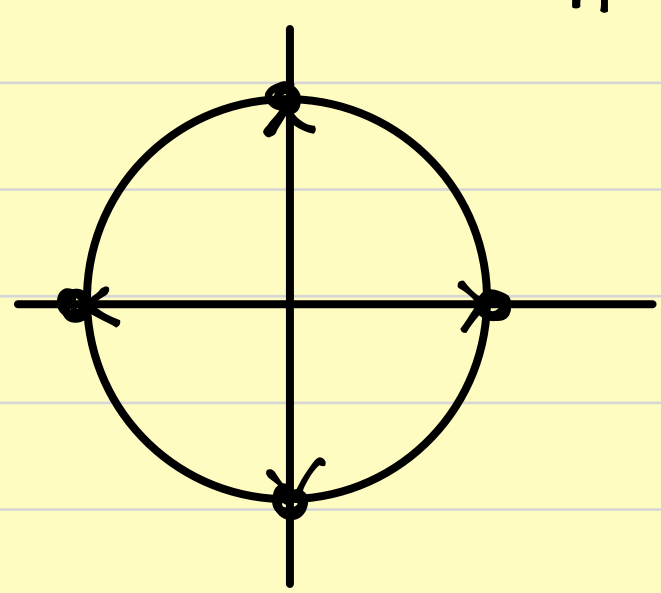
$z_0 = e^{j0} = 1$

$z_1 = e^{j\frac{\pi}{2}} = j$

$z_2 = e^{j\pi} = -1$

$z_3 = e^{j\frac{3\pi}{2}} = -j$

$z_4 = 1$



Now we move to general polynomial equations.

$$P(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$$

A number s_0 is called "root" or "zero" if $P(s_0) = 0$.

$$\boxed{\text{Theorem: A polynomial of degree "n" has exactly "n" complex roots (counting multiplicity).}}$$

First practical case: quadratic equation.

$$as^2 + bs + c = 0 \qquad a \neq 0$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

if $b^2 - 4ac < 0$ then the roots $\in \mathbb{C}$.

example: $s^2 + 2s + 5 = 0 \rightarrow s = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2j$

$s_1 = -1 + 2j \qquad s_2 = -1 - 2j$

We get a complex conjugate pair. $\rightarrow$ OSCILLATORY BEHAVIOUR

Suppose the polynomial has $\mathbb{R}$ coefficients $a, b, c, \dots$
then if $a + jb$ is a root then $a - jb$ is also a root.

example. suppose a polynomial with real coefficients has the root $2 + j \cdot 3$

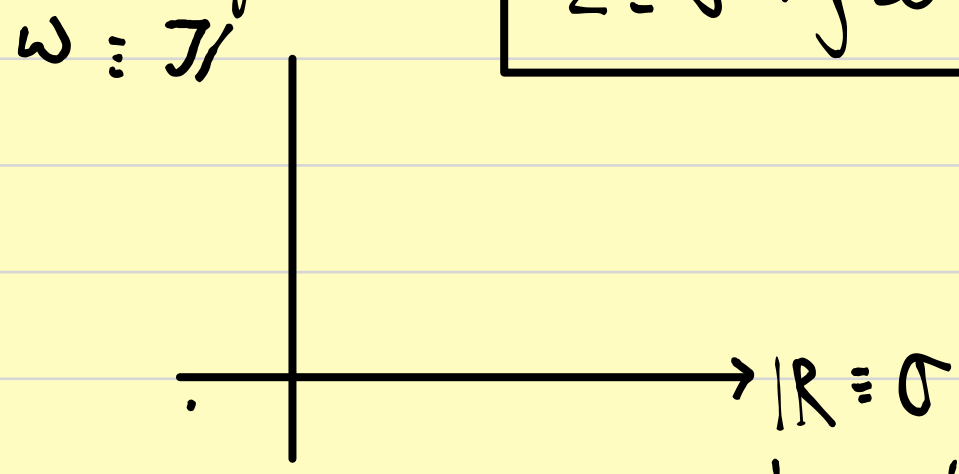
Find the second root and the quadratic factor with real coefficients.

2nd root $\quad 2 - j3$

Quad. Factor: $\left(s-(2+j3)\right)\left(s-(2-j3)\right) =$

$$= (s-2)^2 + 3^2 = s^2 - 4s + 4 + 9 = s^2 - 4s + 13$$

From now on the general complex variable z will have the form $\boxed{z = \sigma + j\omega}$.



Let $s = \sigma + j\omega$ then $e^{st} = e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t}$.

Using Euler's formula: $e^{st} = e^{\sigma t} \cdot \left(\cos(\omega t) + j\sin(\omega t)\right)$

Interpretation is: σ controls GROWTH or DECAY

ω controls OSCILLATORY BEHAVIOUR

If $\sigma < 0$, then $e^{\sigma t}$ decays.

$\sigma > 0$, then $e^{\sigma t}$ grows.

$\omega \neq 0$, then motion oscillates.

How polynomials arise from differential equations:

Consider the diff equation: $y'' + 2y' + 5y = 0$

Try solutions of the form $y(t) = e^{st}$ sc ¢

Then $y'(t) = s e^{st}$

$y''(t) = s^2 e^{st}$

Insert this into the diff eq: $s^2 e^{st} + 2 s e^{st} + 5 e^{st} = 0$

$e^{st} \neq 0 \longrightarrow s^2 + 2s + 5 = 0 \longrightarrow s_0 = -1 + 2j$

$s_1 = -1 - 2j$

EULER $e^{(-1+2j)t} = e^{-t}(\cos 2t + j \sin 2t)$

$e^{(-1-2j)t} = e^{-t}(\cos 2t - j \sin 2t)$

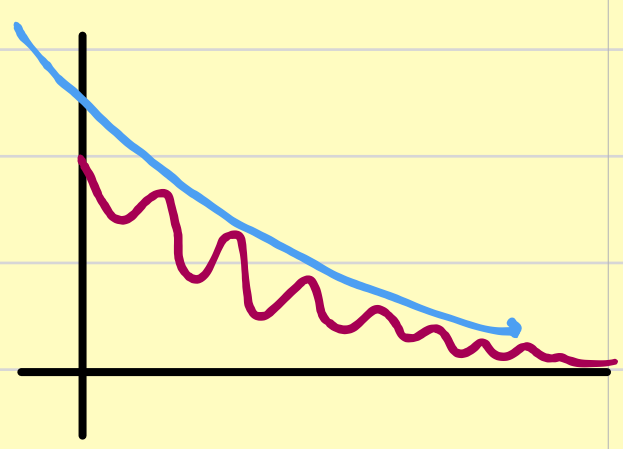
The real-valued solution is a combination (linear) of $e^{-t}\cos 2t$ $e^{-t}\sin 2t$

$$y(t) = e^{-t}(c_1 \cos 2t + c_2 \sin 2t)$$

The real part is -1

$\Downarrow$

Decay

$y'' + 2y' + 5y = 0$

$\longrightarrow$